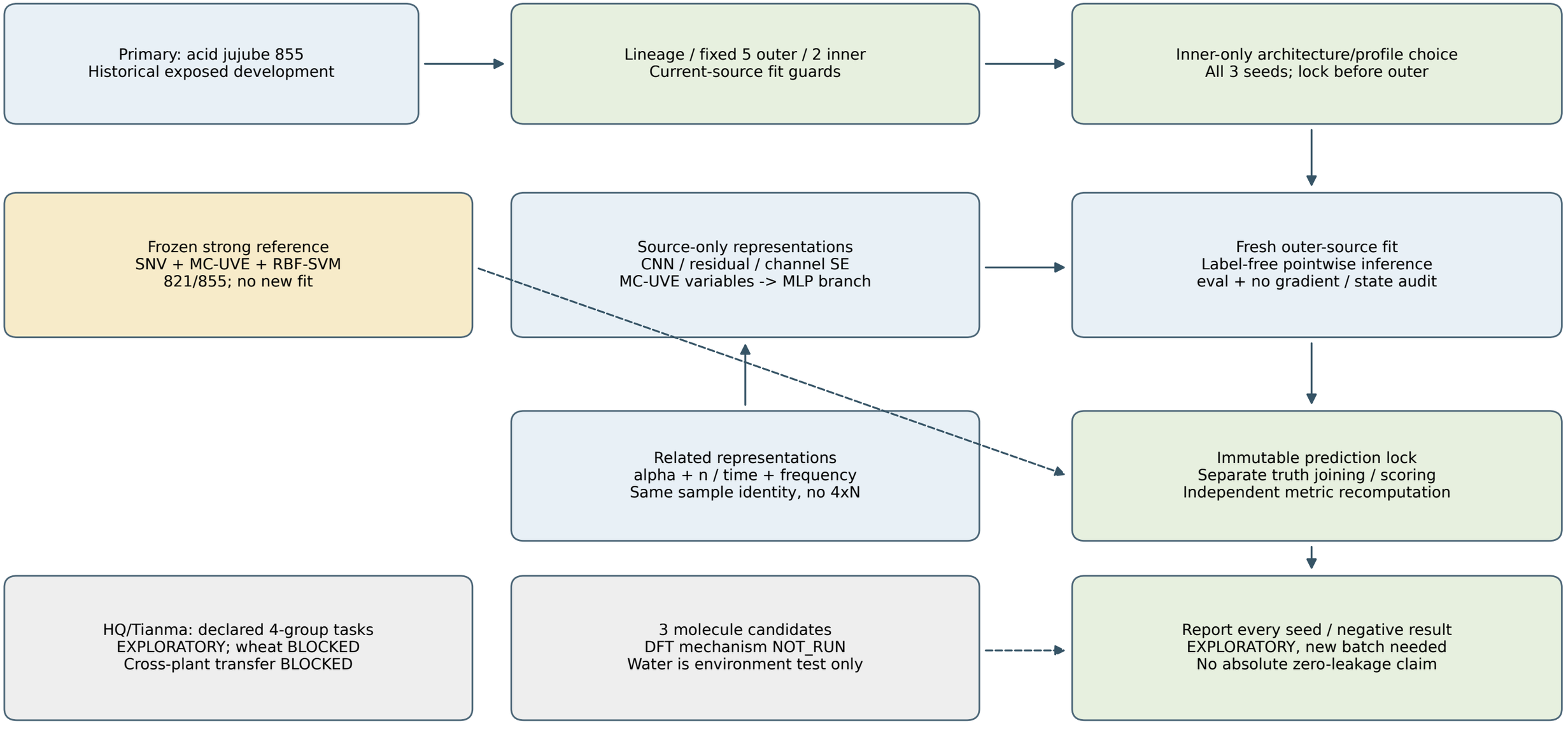Research4 upgrade: preserve acid jujube gains; guarded neural comparison and auxiliary work